



# Network Devices

*Hubs, Switches, Routers, and Access Points*

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2/25/2016



# Networking Devices

- \* Hubs
- \* Switches
- \* Routers
- \* Access Points



Hub



Switches



Router



Wireless  
Access Point

# Motivation for using Networking Devices

- \* Allow a greater number of nodes to be connected to the network.
- \* Extend the distance over which a network can extend.
- \* Localize traffic on the network.
- \* Merge existing networks.
- \* Isolate network problems so that they can be diagnosed more easily.

# Functions of Network Devices

- \* Connect computers as well as other computing devices together in order to share resources.
- \* Separating (connecting) networks or expanding network
  - \* e.g. hubs, routers, switches, access points

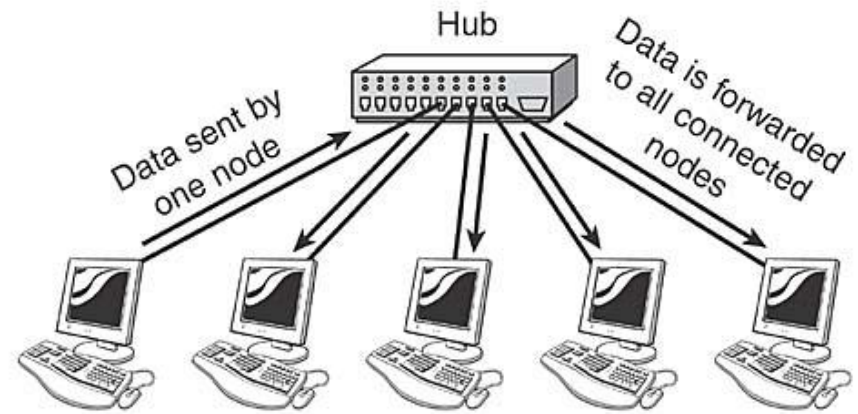


# What is a Hub?

- \* The simplest network device that connects computers together on a twisted-pair network.
- \* Hub works at the OSI physical layer (Layer 1) to regenerate the network's signal and resend them to other segments
- \* Limitations:
  - \* Cannot filter network traffic
  - \* Cannot determine best path
  - \* They are used as network concentration points
  - \* They are really multi-port repeaters

# How does a Hub work?

- \* Receives data packets and passes on all the information it receives to all the other computers connected to the hub.



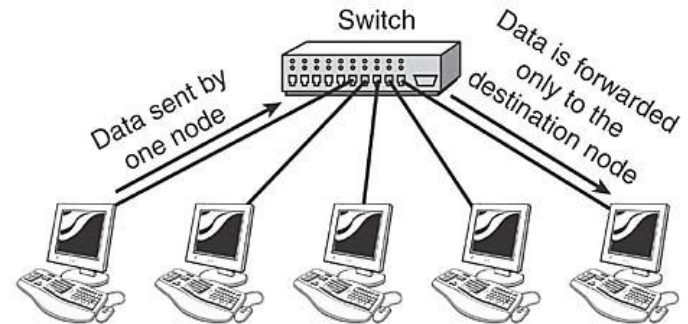


# What is a Switch?

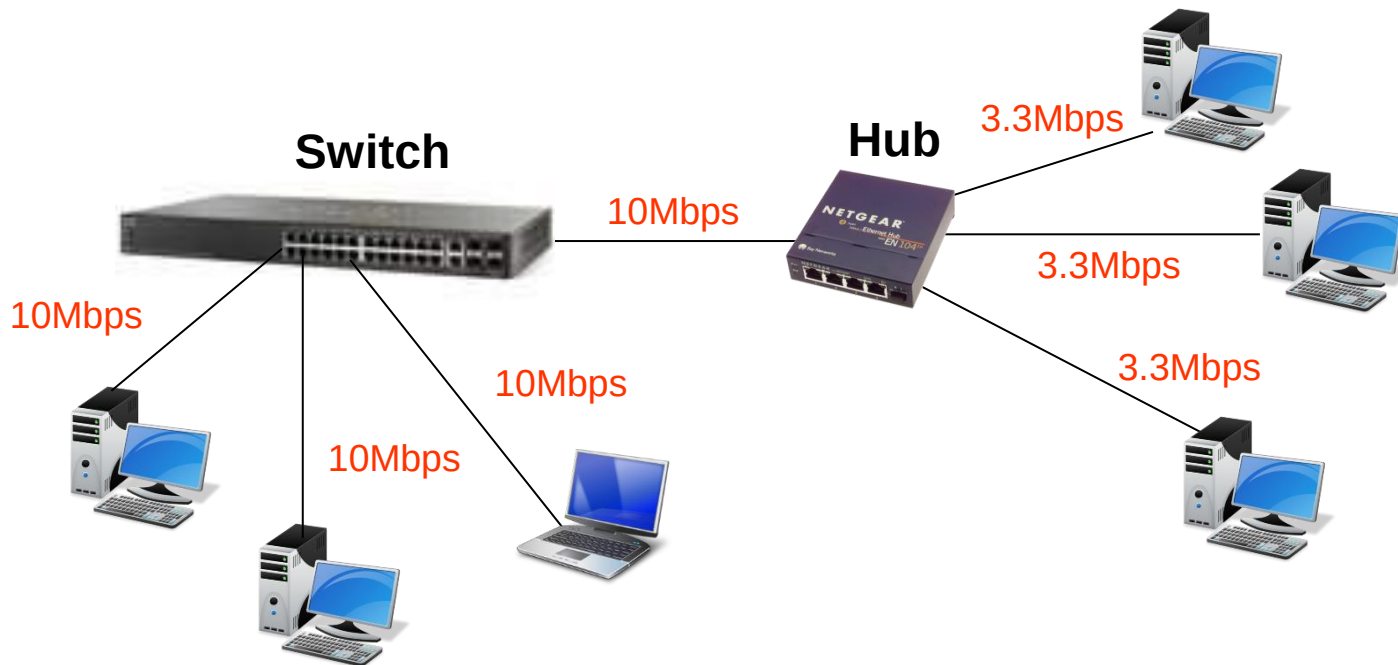
- \* Operates at the OSI physical layer (Layer 2)
- \* Connects and passes packets between two network segments.
- \* Works like a hub but with intelligent and has better performance.
- \* Packets and forwards based on addressing information.
- \* Data travels faster and more secure
- \* Information travels more efficiently through a switch because travels directly to its destination as opposed to being broadcast to all PC's on the network hub.

# How does a Switch work?

- \* Identify the destination of a packet that forward data to its destination by using the MAC Address.
- \* Send information only to the computer that is suppose to receive the information.
- \* Send and retrieve information at the same time which makes sending information faster to retrieve than a hub.



# Typical Switch Network



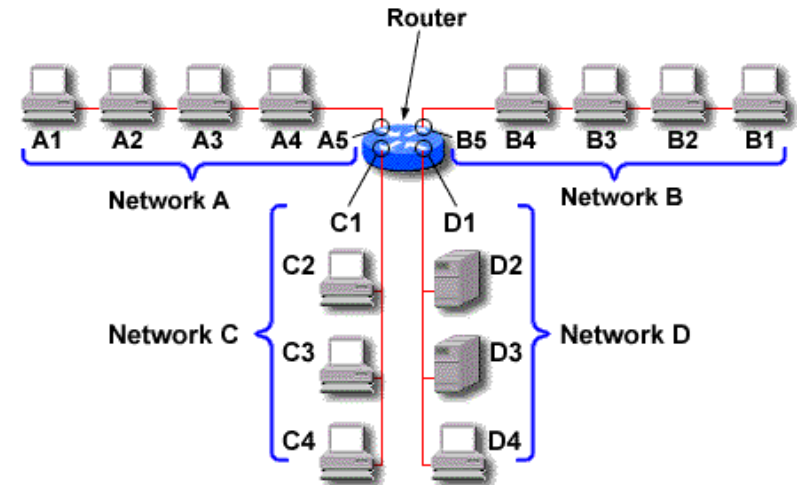


# What is a Router?

- \* Router is another type of internetworking device.
- \* Passes data packets between networks based on the OSI network layer (Layer 3)
- \* Has the ability to make intelligent decisions as to the best path for delivery of data on the network.

# How does a Router work?

- \* As packets are passed from routers to routers, Data Link layer source and destination addresses are stripped off and then recreated
- \* Only packets with known network addresses will be passed - hence reduce traffic
- \* Routers can listen to a network and identify its busiest part
- \* Will select the most cost effective path for transmitting packets



**The router would determine to send the data from network A to network B out its port with the IP address B5.**

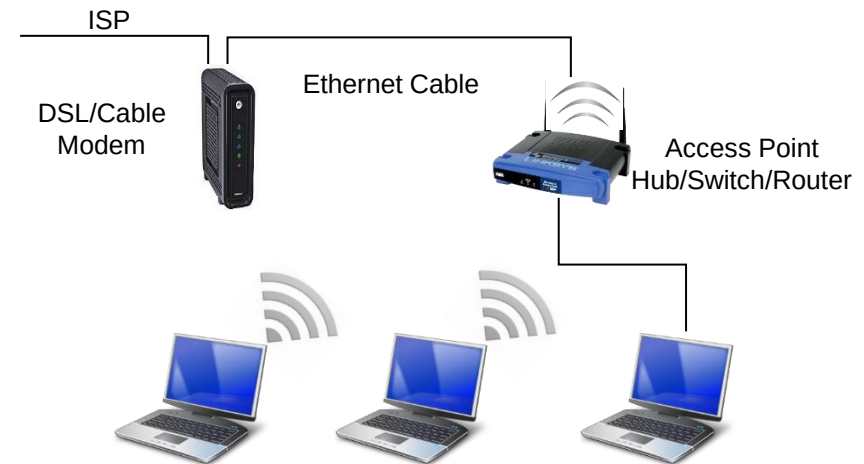


# Wireless Access Point

- \* Connects wireless devices together
- \* Communicates with each other wireless devices
- \* Communicates with a wired network
- \* Most Wireless Access Points today have built-in router, hub/switch, and access point in a single unit.

# How a Wireless Access Point works?

- \* Operates using wireless technology that connects computers without cables
- \* Broadcast wireless signals (Wi-Fi)
- \* Most computers today already have network adapters (Wi-Fi) built into the computer.



# SUMMARY

| Device                                                                                                      | Function                                                   | Key Points                                                                                  |
|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| <b>Hubs</b><br>            | Connects devices on a the network.                         | Connects devices on a twisted-pair network.                                                 |
| <b>Switches</b><br>        | Connects devices on the network.                           | A switch forwards data to its destination by using the MAC address embedded in each packet. |
| <b>Routers</b><br>        | Connects networks together.                                | A router uses the software-configured network address to make forwarding decisions.         |
| <b>Access Points</b><br> | Provides network capabilities to wireless network devices. | A Wireless Access Point (WAP) is often used to connect to a wireless and wired network.     |

